

Name: _____

Class: _____

Date: _____

Lab 8: Variables, Lists & Math

Python Lab · Section 3.1–3.4 Variables, Lists, Math & Strings



Goal

Practice assignment, MOD, lists, and strings in Python — and translate each to AP pseudocode.

Warm-up: Predict

```
nums = [10, 20, 30]
print(nums[0])
print(nums[len(nums) - 1])
print(17 % 5)
```

Predict the output. Write the AP version of each index.

AP Connection

- Python lists start at index 0; AP lists start at 1!
- % in Python is MOD in AP pseudocode.
- Python slicing vs. AP substring.

Build It

Step 1. MOD practice: is a number even? What is its last digit?

```
n = int(input("Number: "))
print("Even?", n % 2 == 0)
print("Last digit:", n % 10)
```

Step 2. Lists: create, change, add, insert, and remove.

```
pets = ["cat", "dog"]
pets.append("fish")      # AP: APPEND
pets.insert(1, "bird")   # AP: INSERT(pets, 2, "bird")
pets.pop(0)              # AP: REMOVE(pets, 1)
print(pets, len(pets))
```

Step 3. Strings: build a username from a first and last name.

```
first = input("First: ")
last = input("Last: ")
user = first[0:3] + last
print(user.lower())
```

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Build It (continued) · Level Up

Build It (continued)

Step 4. Swap two variables using a temp variable, then the Python shortcut.

```
a = 1
b = 2
temp = a
a = b
b = temp
# Python shortcut: a, b = b, a
```

Level Up Challenges

Mild Given a number of days, print the weeks and leftover days.

Medium Reverse a string using a loop (or slicing `::-1`).

Spicy Make a list of 5 numbers and rotate it so the first element moves to the end.

Python → AP Pseudocode

```
pets.append("fish")
pets.insert(1, "bird")
pets.pop(0)
print(pets[0])
```

Write each in AP pseudocode. Watch the indexes!

Exit Ticket

Python index 2 is which AP index? Explain.
